

252-0058-00L Formal Methods and Functional Programming

Semester	Spring Semester 2024
Lecturers	D. Basin, P. Müller
Periodicity	yearly recurring course
Language of instruction	English

◀ Tabs

Courses

Number	Title	Hours	Lecturers
252-0058-00 V	Formal Methods and Functional Programming Findet im HG E 7 mit Videoübertragung ins HG E 3 statt.	4 hrs Tue 10:15-12:00 HG E 3 » 10:15-12:00 HG E 7 » Thu 10:15-12:00 HG E 3 » 10:15-12:00 HG E 7 »	D. Basin, P. Müller
252-0058-00 U	Formal Methods and Functional Programming	2 hrs Tue 14:15-16:00 CAB G 52 » 16:15-18:00 ETZ F 91 » 16:15-18:00 ETZ G 91 » Wed 08:15-10:00 CAB G 52 » 08:15-10:00 ETZ F 91 » 08:15-10:00 HG E 33.5 » 08:15-10:00 LEE C 114 » 16:15-18:00 CHN D 42 » 16:15-18:00 CHN F 46 » 16:15-18:00 HG G 26.5 »	D. Basin, P. Müller

Catalogue data

Abstract	In this course, participants will learn about new ways of specifying, reasoning about, and developing programs and computer systems. The first half will focus on using functional programs to express and reason about computation. The second half presents methods for developing and verifying programs represented as discrete transition systems.
Learning objective	In this course, participants will learn about new ways of specifying, reasoning about, and developing programs and computer systems. Our objective is to help students raise their level of abstraction in modeling and implementing systems.
Content	The first part of the course will focus on designing and reasoning about functional programs. Functional programs are mathematical expressions that are evaluated and reasoned about much like ordinary mathematical functions. As a result, these expressions are simple to analyze and compose to implement large-scale programs. We will cover the mathematical foundations of functional programming, the lambda calculus, as well as higher-order programming, typing, and proofs of correctness.

The second part of the course will focus on deductive and algorithmic validation of programs modeled as transition systems. As an example of deductive verification, students will learn how to formalize the semantics of imperative programming languages and how to use a formal semantics to prove properties of languages and programs. As an example of algorithmic validation, the course will introduce model checking and apply it to programs and program designs.

Competencies 	Subject-specific Competencies	Concepts and Theories	assessed
		Techniques and Technologies	assessed
	Method-specific Competencies	Analytical Competencies	assessed
		Problem-solving	assessed
	Personal Competencies	Creative Thinking	fostered
		Self-direction and Self-management	assessed

Performance assessment

Performance assessment information (valid until the course unit is held again)	
► Performance assessment as a semester course	
ECTS credits	7 credits
Examiners	P. Müller, D. Basin
Type	session examination
Language of examination	English
Repetition	The performance assessment is only offered in the session after the course unit. Repetition only possible after re-enrolling.
Mode of examination	written 180 minutes
Additional information on mode of examination	We offer 2 graded midterms, in the first and second half. Each midterm will be 30 minutes and act as a bonus, that is, may improve the grade, but will not make it worse. In the positive case, each midterm will count 10% of the total grade.
Written aids	None

This information can be updated until the beginning of the semester; information on the examination timetable is binding.

Learning materials

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Main link	Information
Only public learning materials are listed.	

Groups

No information on groups available.

Restrictions

There are no additional restrictions for the registration.
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Offered in

Programme	Section	Type
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